

Predictors of Opioid Misuse Disorder in the US

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Abstract

Background: Opioid use disorder and pain reliever misuse remain major public health concerns in the United States.

Research Question: What factors are important for predicting prevalence rate of opioid use disorder (OUD)?

Methods: We used state-level National Survey on Drug Use and Health data from 2018–2019 and 2021–2024. After exploratory data analysis and Box-Cox transformation of numeric predictors, we compared a full linear regression model, a stepwise AIC model, and a LASSO model. Model selection was based on 10-fold cross-validated RMSE.

Results: The final LASSO model identified educational attainment, poverty, age composition, employment, Medicaid coverage, and racial composition as the strongest predictors of pain reliever misuse. Some college education, the share of children aged 12 to 17, full-time employment, and Medicaid coverage were positively associated with misuse, while higher shares of people with a college degree, less than a high school education, incomes above two times the federal poverty level, and non-Hispanic white and Black residents were negatively associated with misuse.

Conclusion: The strongest state-level predictors of pain reliever misuse were educational attainment, poverty, and age composition, with additional contributions from employment, Medicaid coverage, and racial composition. Because the data are cross-sectional and aggregated at the state level, the results should be interpreted as associations rather than causal effects.

Research Question:

What factors are important for predicting prevalence rate of opioid use disorder (OUD)?

Introduction

Opioid addiction is a public health crisis in the US that has affected countless lives.

We obtained data on the prevalence of pain reliever misuse in each state from the National Survey on Drug Use and Health (NSDUH) from the [Substance and Mental Health Services Administration](#) (SAMHSA).

In this presentation, we describe our data analysis and findings of our investigation of the factors predicting the prevalence rate of OUD across US states.

Methods

We used the following steps and tools to answer our research question:

Data Acquisition

Datasets (CSV)

- National Survey on Drug Use and Health by year
- Merged annual opioid misuse and predictor variable data by state

Data Preparation

- Removed irrelevant columns and rows
- Converted data types (eg, string to date)
- Filtered rows for past-year opioid misuse prevalence
- Applied Box-Cox transformations to numeric predictors to normalize distributions
- Split data 70:30 into training and test sets

Modeling

- Checked assumptions for regression
- Developed 3 models (Full, Stepwise Selection, & LASSO regression)
- Performed Hyperparameter Tuning to tune models
- Use 10-fold Cross-Validation to select the model with the lowest RMSE

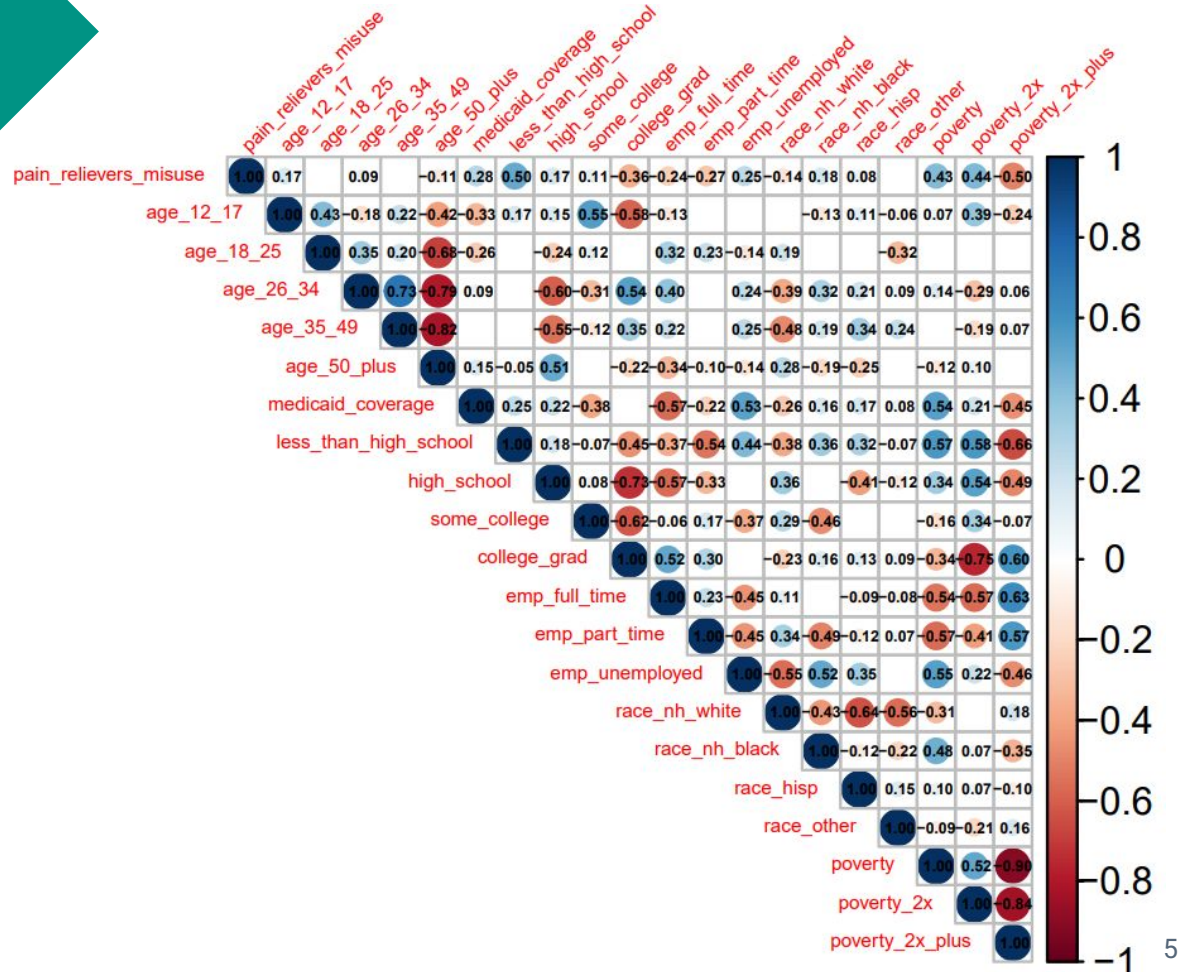
OUD, opioid use disorder.

All data and code are available at: <https://github.com/naomibuell/DATA621/tree/main/final-project>.

Exploratory Data Analysis

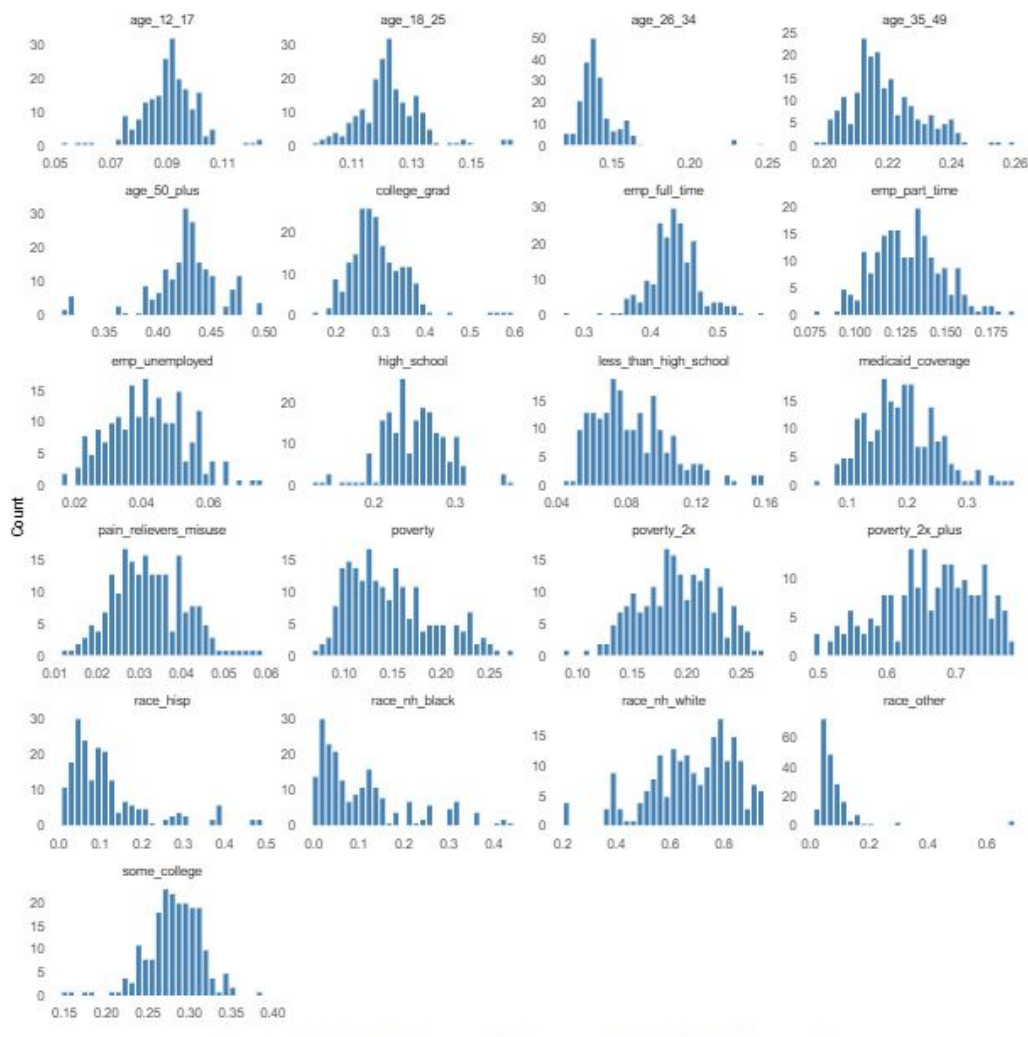
- We visualized the correlations between predictors and the outcome variable.
- OUD (pain_relievers_misuse in top row) was **positively correlated** with:
 - Having less than a high school education
 - Higher rates of poverty
 - Medicaid coverage
- OUD was **negatively correlated** with:
 - College graduation rate
 - Employment rates

OUD, opioid use disorder.



Exploratory Data Analysis

- We visualized the distributions of numeric variables.
- Most variables appear normally distributed.
- Some variables (e.g., race variables) appear to be non-normal, which may need to be box-cox transformed in the following data preparation step.



Data Preparation

- Box-Cox transformations were used to normalize distributions of numeric predictors
- No data were missing, so imputation was not necessary
- No additional features were engineered to facilitate comparison of results with a previous study
- 70:30 train:test data split

Variable(s)	Optimal λ^a	Transformation
age_50_plus, poverty_2x_plus, race_nh_white, some_college	2	Squared
age_12_17, emp_full_time, high_school, poverty_2x	1	None
emp_unemployed, medicaid_coverage	0.5	Square root
poverty, race_hisp, race_nh_black	0	Natural log
college_grad, race_other, less_than_high_school	-0.5	Reciprocal square root
age_18_25	-1	Inverse
age_26_34, age_35_49	-2	Inverse squared

^a For Box-Cox transformation. λ values rounded to the nearest “common” transformation.

Assumptions

This section outlines key regression assumptions, including linearity, independent state-year observations, constant variance with approximately normal residuals, and expected multicollinearity among predictors.

- Linearity between predictors and outcome
- Independent state-year observations
- Constant variance & approximately normal residuals
- Some multicollinearity expected

01 OLS-based model (Full Regression & Stepwise AIC)

Rely on standard OLS assumptions: linearity, independence, homoscedasticity, normal residuals

Observations assumed independent across states and years

02 Multicollinearity Considerations

Present due to compositional predictors (e.g., race, age, education proportions summing to 1)

Expected correlation among demographic variables

03 LASSO Model

Does not require strict normality of residuals

Handles multicollinearity using L1 regularization

Shrinks less important predictors toward zero, improving model stability with many correlated variables

Building Models

We evaluate regression techniques to balance model fit with predictive accuracy using a multi-stage validation approach.

- Full MLR (Baseline)
- Stepwise AIC Selection
- LASSO Regularization
- 10-fold Cross-Validation

01

Full Multiple Linear Regression

Constructed using all 20+ predictors to establish a performance ceiling for R^2 and residual standard error.

02

Stepwise Variable Selection

Iteratively adds/removes variables to minimize AIC, penalizing complexity to favor simpler explanations.

03

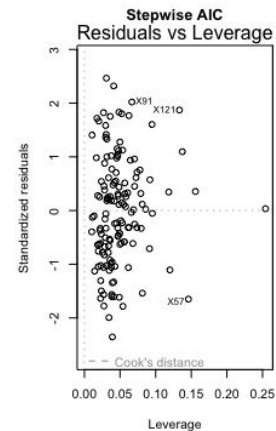
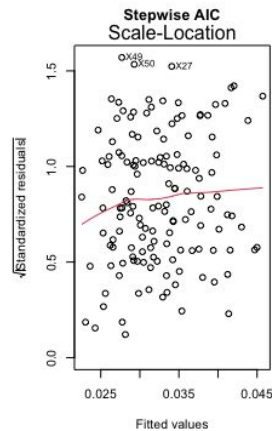
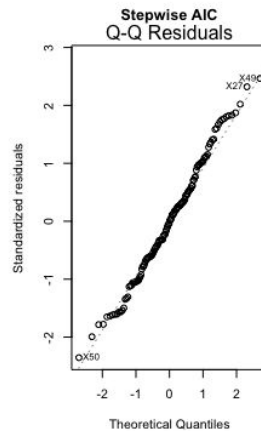
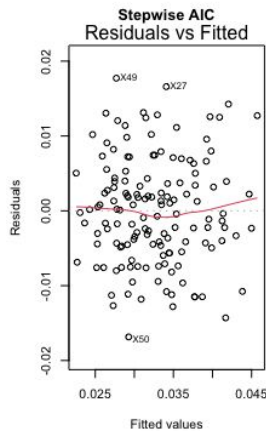
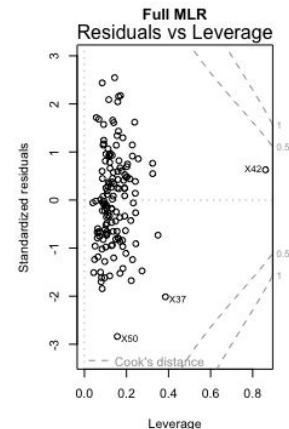
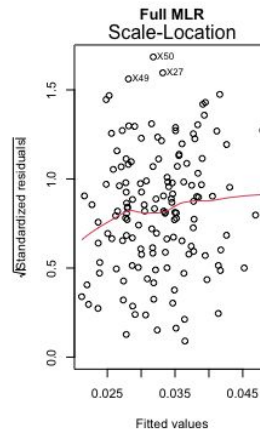
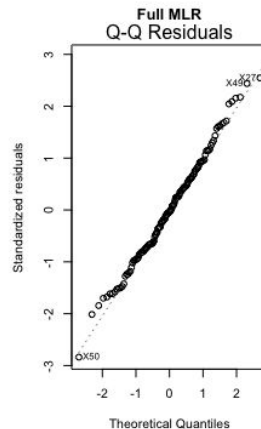
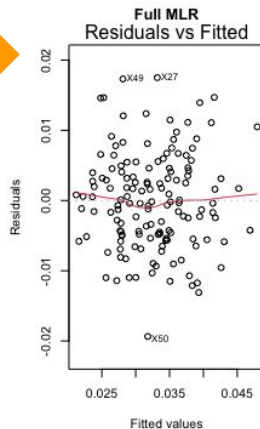
LASSO Regression

Addresses multicollinearity by shrinking insignificant coefficients to zero via hyperparameter tuning (λ).

```
# Define 10-fold CV & Train Full MLR
train_control <- trainControl(method = "cv", number = 10)
model_full <- train(pain_relievers_misuse ~ .,
data = oud_train_split, method = "lm",
trControl = train_control)
```

Model Diagnostics

- **Linearity:** Residual–fitted plots show no strong curvature, supporting a linear relationship.
- **Normality:** Q–Q plots align closely with the diagonal, with minor tail deviations and one consistent outlier (X50).
- **Homoscedasticity:** Residual variance is mostly stable; slight improvement in the Stepwise AIC model.
- **Influence:** One high-leverage point (X42) in Full MLR is reduced in Stepwise AIC, indicating improved stability.



Model Diagnostics

- **Multicollinearity (VIF results):** Strong multicollinearity observed across many predictors.
- **Highest VIFs:** age_50_plus (236.6) and poverty_2x_plus (196.4), with several others > 10.
- **Cause:** Expected due to compositional variables (age, race, education, poverty proportions summing to 1).
- **Lower VIF variables:** Employment variables, race_nh_black, race_hisp, and medicaid_coverage.
- **Implication:** Supports use of LASSO, which reduces multicollinearity via regularization and variable selection.

Predictor	VIF
age_50_plus	236.58
poverty_2x_plus	196.41
poverty	71.07
poverty_2x	46.41
age_26_34	39.08
age_35_49	39.08
college_grad	37.49
age_18_25	32.04
high_school	22.96
age_12_17	15.83
less_than_high_school	13.54
some_college	12.24
race_nh_white	10.34
race_other	5.39
race_hisp	4.83
emp_full_time	4.61
race_nh_black	3.94
medicaid_coverage	3.30
emp_part_time	3.24
emp_unemployed	3.10

Model Selection Results

To identify the most reliable model for predicting state-level opioid misuse prevalence, we compared three different regression approaches:

- Full Multiple Linear Regression
- Stepwise AIC Regression
- LASSO Regression

All models were evaluated using **10-Fold Cross Validation** and compared using **RMSE**.

01

Why We Compared Multiple Models

Full Model captures all available predictors but may overfit
Stepwise AIC improves interpretability by reducing unnecessary variables
LASSO handles multicollinearity and performs automatic feature selection

02

Goal: Find the best balance between

- Accuracy
- Stability
- Interpretability
- Generalization Performance

03

Key Findings

- Full Model showed strong fit but high complexity
- Stepwise AIC reduced model complexity while maintaining performance
- LASSO achieved the strongest overall predictive performance and stability

04

Final Selected Model: LASSO Regression

- ✓ Lowest Cross-Validated RMSE
- ✓ Reduced Overfitting Risk
- ✓ Better Performance on Unseen Data
- ✓ Most Reliable Model for High-Dimensional Public Health Data

Lasso Model Results: Important Predictors

- **Educational attainment strongly predicted pain reliever misuse:** higher OUD when folks didn't complete college or high school.
- **Economic disadvantage increased misuse:** higher OUD in states with more residents near or below 200% FPL, lower rates in higher-income populations.
- A higher **youth population** share (ages 12-17) also positively correlated with misuse.
- **Employment** and **Medicaid** coverage were positively associated, potentially reflecting healthcare access and vulnerability.
- State **racial composition** also showed associations.

FPL, federal poverty level.

Predictor	LASSO Coefficient	LASSO Importance
some_college	0.084	100.000
poverty_2x_plus	-0.029	34.583
age_12_17	0.012	14.430
emp_full_time	0.010	11.425
medicaid_coverage	0.007	8.143
poverty_2x	0.005	5.902
less_than_high_school	0.004	5.300
college_grad	-0.002	2.239
race_nh_white	-0.001	1.624
race_nh_black	-0.001	1.279

Conclusions

- Overall, the LASSO model identified low levels of education, poverty, more children, employment, Medicaid coverage, and race as key correlates of state-level variation in pain reliever misuse
- Limitations:
 - State-level data cannot identify individual-level risk factors
 - Cross-sectional design prevents causal conclusions
 - Important predictors may be missing from the model

References and Links

1. National Survey on Drug Use and Health data
https://datatools.samhsa.gov/nsduh/2019/nsduh-2018-2019-rd02yr/crosstab?row=PNRNMYR&column=STUSAB&weight=DASWT_1
2. Link to project code:
<https://github.com/naomibuell/DATA621/tree/main/final-project>